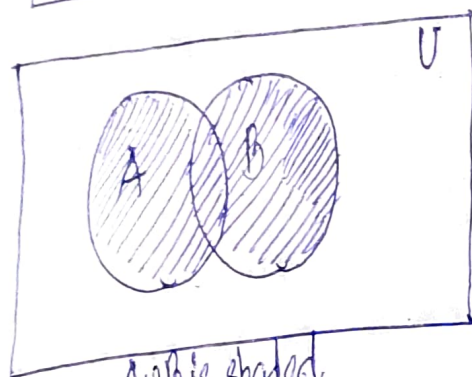


2.2 Set Operations

2.2.1 Introduction

Definition 1: Let A and B be sets. The **union** of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



$A \cup B$ is shaded

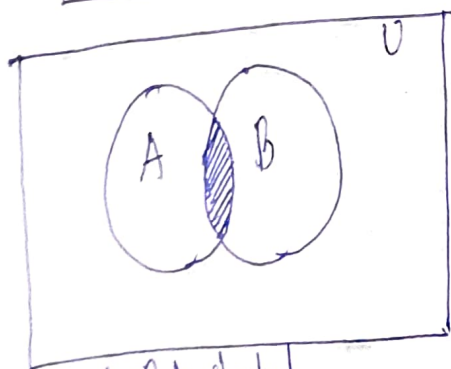
Figure 1: Venn diagram of the union of A and B

Example 1: The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$: $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$ □

Example 2: The union of the set of all computer science majors at your school and the set of all mathematics majors at your school is the set of all students at your school who are majoring either in mathematics or in computer science (or in both). □

Definition 2: Let A and B be sets. The **intersection** of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



$A \cap B$ is shaded

Figure 2: Venn diagram of the intersection of A and B

Example 3: The intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$: $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$ □

Example 4: The intersection of the set of all computer science majors at your school and the set of all mathematics majors is the set of all students who are joint majors in both maths and CS □

Definition 3: Two sets are called **disjoint** if their intersection is the empty set.